

VIRRAT AIJANNEVA, Finland

WMO: 027350

Lat: 62.3278N

Lon: 23.5422E

Elev: 140

StdP: 99.66

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.5	(c) -23.2	(d) -29.0	(e) 0.3	(f) -26.5	(g) -25.5	(h) 0.4	(i) -23.1	(j) 9.3	(k) -2.8	(l) 8.4	(m) -2.9	(n) 0.4	(o) 160	(p) 0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	26.7	18.4	24.7	17.4	22.8	16.5	19.8	24.4	18.6	22.8	17.4	21.4	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	21.6	16.9	12.2	20.4	15.7	11.3	19.1	57.2	24.5	53.2	22.6	49.6	21.5	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 7.2	(c) 6.4	(e) -29.2	(f) 28.4	(g) 3.0	(h) 2.9	(i) -31.4	(j) 30.5	(k) -33.1	(l) 32.2	(m) -34.8	(n) 33.9	(o) -37.0	(p) 36.0		
(a) 8.2	(b) 7.2	(c) 6.4	(e) -29.4	(f) 21.2	(g) 3.0	(h) 2.0	(i) -31.5	(j) 22.6	(k) -33.2	(l) 23.8	(m) -34.9	(n) 24.9	(o) -37.0	(p) 26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-8.1	-6.3	-3.3	2.7	9.1	12.8	15.8	14.0	9.8	3.4	-0.1	-4.4
	DBStd	9.33	6.91	6.92	5.41	3.63	4.42	3.47	3.34	2.95	3.38	3.81	4.71	6.27
	HDD10.0	2731	561	456	413	218	70	11	1	3	44	207	302	445
	HDD18.3	5312	819	689	671	468	288	169	94	137	257	463	552	703
	CDD10.0	484	0	0	0	1	41	95	180	129	37	2	0	0
	CDD18.3	23	0	0	0	0	1	3	15	4	0	0	0	0
	CDH23.3	325	0	0	0	0	29	57	191	47	0	0	0	0
	CDH26.7	51	0	0	0	0	1	6	34	9	0	0	0	0
(14) Wind	WSAvg	3.0	3.1	3.3	3.5	3.1	3.1	3.1	2.5	2.7	2.8	3.0	3.1	3.3
(15) Precipitation	PrecAvg	556	39	29	29	28	40	62	76	65	53	53	43	40
	PrecMax	744	81	65	49	58	81	106	136	126	116	112	96	74
	PrecMin	422	10	1	3	2	5	13	12	16	13	9	2	14
	PrecStd	76	19	16	11	18	23	25	33	27	33	29	21	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	5.4	9.3	18.7	26.3	27.5	29.6	28.1	20.8	13.8	10.4	6.9
		MCWB	2.2	2.9	4.3	9.7	15.3	18.0	20.2	20.1	15.7	11.5	9.2	6.2
	2%	DB	2.0	3.7	6.3	14.7	23.8	25.0	27.5	24.2	18.5	11.7	8.2	4.4
		MCWB	1.1	2.2	3.2	8.3	14.4	17.2	19.5	17.7	14.5	10.4	7.4	3.6
	5%	DB	1.0	2.5	4.7	11.5	21.6	22.4	25.5	22.3	16.9	10.6	6.9	3.0
		MCWB	0.5	1.7	2.3	6.5	13.5	15.8	18.3	16.4	13.5	9.4	6.4	2.3
	10%	DB	0.2	1.2	3.2	9.3	18.7	20.4	23.5	20.4	15.5	9.2	5.4	2.0
		MCWB	-0.2	0.7	1.3	5.4	12.0	14.7	17.7	15.5	13.1	8.2	5.0	1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	3.2	4.8	10.1	16.8	19.9	22.0	20.7	16.5	12.0	9.4	6.3
		MCDB	3.1	4.4	7.2	16.8	23.2	24.6	27.0	26.2	19.4	13.2	10.2	6.7
	2%	WB	1.3	2.4	3.6	8.7	15.5	18.1	20.4	18.5	15.3	10.9	7.6	3.8
		MCDB	1.7	3.4	5.7	13.5	21.3	22.9	25.2	22.5	17.3	11.6	8.2	4.4
	5%	WB	0.6	1.7	2.6	7.1	14.2	16.8	19.4	17.3	14.4	9.7	6.3	2.4
		MCDB	0.9	2.5	4.2	10.8	20.0	21.2	23.8	20.9	16.2	10.4	6.8	2.8
	10%	WB	-0.1	0.8	1.6	5.8	12.8	15.5	18.3	16.3	13.5	8.2	5.1	1.5
		MCDB	0.3	1.2	2.9	8.7	18.5	19.5	22.4	19.2	15.1	9.1	5.4	1.8
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.5	9.0	9.4	11.6	10.7	11.0	11.0	8.9	6.0	3.9	4.8
		MCDBR	3.8	5.5	8.4	14.5	17.5	15.0	14.2	14.5	11.5	7.5	4.8	4.6
		MCWBR	3.6	4.6	5.8	8.8	8.9	8.2	7.1	8.2	7.9	6.1	4.5	4.4
	5% WB	MCDWR	3.7	5.0	7.4	12.4	15.2	12.6	12.6	11.8	9.6	6.7	4.5	4.6
		MCWBR	3.7	4.2	5.5	8.0	8.2	7.3	7.0	7.0	6.9	6.0	4.3	4.5
(40) Clear-Sky Solar Irradiance	taub		0.237	0.288	0.314	0.353	0.349	0.348	0.371	0.356	0.332	0.315	0.258	0.353
	taud		2.061	2.164	2.240	2.355	2.444	2.469	2.423	2.456	2.518	2.408	2.157	2.615
	Ebn at Noon		428	657	788	833	870	877	842	825	781	639	400	267
	Edh at Noon		34	66	89	99	98	98	100	89	70	53	31	21
(44) All-Sky Solar Radiation	RadAvg		0.20	0.76	2.27	3.68	4.96	5.34	5.03	3.86	2.29	0.93	0.24	0.10
	RadStd		0.03	0.17	0.24	0.45	0.53	0.46	0.50	0.41	0.27	0.19	0.04	0.01

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-131	N/A	N/A	N/A

Nomenclature: See separate page